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Check Your Understanding: Biological Treatments for Depression

Practice Assignment • 6 min

Due Sep 16, 11:59 PM IST

Your grade: 100%

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Instructions

1. **Which of the following works to inhibit the breakdown of neurotransmitters in the presynaptic neuron?** 1 / 1 point

● MAOIs
○ TCAs
○ SNRIs
○ SSRIs

✓ **Nice work**

MAOIs inhibit the enzyme monoamine oxidase from breaking down neurotransmitters in the presynaptic neuron. SSRIs, SNRIs, and TCAs, however, all work in various ways to block the reuptake of neurotransmitters.

2. **Which of the following is a reason that SSRIs are prescribed more commonly than MAOIs?** 1 / 1 point

● They have fewer side effects
○ They affect more neurotransmitter systems
○ They provide more long-lasting effects
○ They require a smaller dose

✓ **Nice work**

Compared to MAOIs, both SSRIs and SNRIs have fewer side effects and more targeted physiological mechanisms (e.g., target specific neurotransmitters).

3. **Lithium is used as a first line treatment for bipolar disorder because** 1 / 1 point

○ it focuses only on depressive episodes.
○ it does not include any side effects.
● it treats both manic and depressive episodes.
○ the physiological mechanisms are clear.

✓ **Nice work**

Lithium remains the most effective medication for long term treatment of bipolar disorder because it targets both manic and depressive episodes. However, the physiological mechanisms of lithium are unclear and there are concerns about serious side effects.

4. **Repetitive transcranial magnetic stimulation (rTMS) includes** 1 / 1 point

○ destroying or removing small amounts of brain matter.
○ a surgical procedure for the installation of the repetitive transcranial magnetic neurostimulator.
○ targeted and direct electrical stimulation of specific brain areas.
● targeted, but indirect electrical stimulation to specific brain areas.

✓ **Nice work**

Repetitive transcranial magnetic stimulation is an outpatient procedure and no surgery is required. A magnetic coil is placed just above the scalp of the target brain region. The electromagnetic field generated at the coil sends pulses to stimulate the targeted brain region, however there is no direct stimulation of brain areas.

5. **Vagus nerve stimulation employs a pulse-generator that sends signals to the vagus nerve.** 1 / 1 point

● True
○ False

✓ **Nice work**

True. Vagus nerve stimulation stimulates the brain indirectly via the vagus nerve. A surgically implanted pulse generator sends electrical impulses to an electrode located beside the vagus nerve. Stimulation of the vagus nerve cascades and activates the locus coeruleus, a brain region that is networked and interconnected to multiple brain regions including regions responsible for mood regulation.

6. **Deep brain stimulation includes** 1 / 1 point

○ electrical stimulation of a large region of the brain.
○ indirect brain stimulation through nerves that integrate with various brain regions.
○ non-invasive methods to target subcortical areas of the brain.
● implantation of electrodes for specific, targeted brain stimulation.

✓ **Nice work**

Rather than a broad surge of impulses through the brain, the surgical implantation of electrodes in subcortical structures of the brain allows for specific and targeted brain stimulation. These embedded electrodes are powered by a surgically implanted pulse-generating device.

^ **See feedback**

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